

BioinformaticsTasks & Instructions — CodeAlpha

Bioinformatics — Internship Overview

This internship program provides practical exposure to bioinformatics, an interdisciplinary field combining biology, computer science, and data analysis. CodeAlpha is dedicated to building future-ready professionals skilled in computational biology and genomic research. The internship equips students with the ability to analyze biological sequences, perform molecular modeling and use computational tools for biological data interpretation. Interns will gain hands-on experience in sequence alignment, protein structure prediction and database management. Through guided mentorship and real-world projects, participants will develop essential skills for careers in biotechnology, healthcare and research.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete file to **Google drive**:
CodeAlpha_ProjectName
 4. Post a **video explanation** of your project on LinkedIn.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 2 or 3 out of the 4 tasks** listed below (from your domain).
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Bioinformatics Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: DNA/Protein Sequence Analysis

Download a DNA or protein sequence from **NCBI** or **UniProt**.

- Perform **BLAST analysis** to find homologous sequences.
 - Document results (similarity, identity, alignment score).
 Submit screenshots + analysis report (2–3 pages).
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✓ TASK 2: Multiple Sequence Alignment

Choose at least **5 protein sequences** of the same family.

- Perform **Multiple Sequence Alignment (MSA)** using **Clustal Omega/MUSCLE**.
 - Highlight conserved regions and motifs.
👉 Submit alignment results with interpretation.
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✓ TASK 3: Protein Structure Prediction

Select one protein sequence from UniProt.

- Use tools like **SWISS-MODEL** or **AlphaFold DB** to predict its 3D structure.
 - Visualize the structure in **PyMOL/Chimera**.
👉 Submit structure images + short report on structural features.
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✓ TASK 4: Mini Project – Bioinformatics Database/Tool Study

Prepare a **mini-project report (5–6 pages)** on one of the following:

- Comparative study of bioinformatics databases (NCBI, UniProt, PDB).
 - Role of bioinformatics in drug discovery.
 - Applications of machine learning in bioinformatics.
👉 Submit in PDF format with proper references.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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